DASK-Net: A Lightweight Dual-Attention Selective Kernel Network for Efficient Dense Prediction in Remote Sensing Imagery

Furkat Sultonov, Sangseok Yun, *Member, IEEE*, and Jae-Mo Kang, *Member, IEEE*

***Abstract*— The precise and efficient extraction of buildings and road networks from aerial imagery is crucial for remote sensing applications, such as map building, urban development, and autonomous driving guidance systems. However, accurately extracting buildings and roads from remote sensing imagery is challenging due to varying resolutions, object scale variation, and diverse appearances of buildings and roads. While sophisticated models achieve high accuracy, their computational demands limit practical use on resource-constrained devices. Conversely, mainstream lightweight models often fail to generate high-quality segmentation maps for remote sensing data. To address these challenges, we introduce a dual-attention selective kernel network (DASK-Net), a novel lightweight architecture for efficient pixel-wise dense prediction. DASK-Net's core features a dual-attention selective kernel (DASK) module that integrates multiscale feature extraction with adaptive receptive fields and dual-attention mechanisms. This design captures diverse scales and orientations of features while focusing on salient input aspects. We conducted experiments on the Massachusetts roads, DeepGlobe, and WHU building datasets, comparing DASK-Net with numerous methods. The results demonstrate that DASK-Net outperforms these networks while significantly reducing computational complexity. With only 0.48M parameters, DASK-Net achieves an 89.54% intersection over union (IoU) and a 94.48% *F*1 score on the WHU building dataset, setting new performance standards for lightweight methods.**

***Index Terms*— Building footprint extraction, channel attention, remote sensing imagery, road extraction, selective kernel (SK) networks, semantic segmentation, spatial attention.**

I. Introduction

IN THE field of aerial imagery analysis, the accurate and efficient extraction of road networks and buildings remains a critical area of study due to a wide range of applications, such as map building [1], [2], urban planning [3], [4], [5], and autonomous driving systems. However, distinctive features inherent in remote sensing data, such as varying image resolutions, objects of different sizes, and diverse geometries of buildings and road networks, present remarkable challenges [6]. Traditional methods for building [7], [8] and road extraction [9], [10], often based on hand-crafted features, have shown limited success in addressing these challenges. With the advent of deep learning (DL), substantial improvements have been achieved in remote sensing image segmentation tasks [11], [12], [13], [14].

BuildFormer [15] exploits the vision transformer (ViT) [16] architecture to extract long-range dependencies, enabling the precise extraction of buildings effectively. The ViT component in BuildFormer divides the input image into patches and treats these patches as a sequence, applying self-attention mechanisms across all pairs of patches. This self-attention allows the model to directly compute relationships between any two positions in the image, regardless of their spatial distance, thus efficiently capturing global context and long-range dependencies.

Although methods that capture long-range dependencies have enhanced the understanding of broader spatial relationships in remote sensing imagery, they still fail to detect smaller objects and fine details effectively. This limitation is particularly problematic in remote sensing applications where accurately delineating small-scale features is crucial. The inherent variation in object sizes within satellite and aerial imagery poses a substantial challenge, as models must simultaneously process global context and local, fine-grained information. To address this, multiscale aggregation fully convolutional networks (MA-FCNs) [14] leverages a feature pyramid network (FPN) to extract multiscale features and introduces a polygon regularization algorithm to refine the segmentation maps generated by FCN. Additionally, multiple attending path neural network (MAP-Net) [1] fuses features extracted from multiple paths and applies channel attention and spatial pooling enhancement operations to capture global dependencies for continuous building footprints. Although these methods achieve promising results, they are computationally intensive, limiting their applicability in resource-constrained environments such as unmanned aerial vehicles (UAVs).

To address this issue, several lightweight networks have been developed for remote sensing image segmentation. RSR-Net [17] fuses shallow and deep features via squeeze-and-excitation to assign channel weights and reduce noise, achieving competitive intersection-over-union (IoU) scores with minimal parameters. However, it lacks multiscale feature extraction capability, limiting its adaptability to varying

Received 17 March 2025; revised 18 April 2025; accepted 28 April 2025. Date of publication 1 May 2025; date of current version 20 May 2025. This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean Government [Ministry of Science and ICT (MSIT)] under Grant RS-2025-00559998. *(Corresponding authors: Sangseok Yun; Jae-Mo Kang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3566097

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building sizes, and relies solely on channel attention without spatial attention, compromising segmentation refinement.

Multiscale level network (MSL-Net) [18] employs depthwise separable convolutions within an inverted residual structure with group normalization, incorporating atrous spatial pyramid pooling (ASPP) for multiscale feature capture and deformable convolution for irregular building shapes. Similarly, LRAD-Net [19] combines a regular networks (RegNets) backbone with a multiscale depthwise separable ASPP structure, employing squeeze-and-excitation mechanisms for channel weight redistribution and lightweight residual blocks with polarized self-attention for efficient feature integration.

Despite their state-of-the-art performance, both MSL-Net [18] and LRAD-Net [19] are limited by their dependence on computationally expensive ASPP/depthwise separable atrous spatial pyramid pooling (DSASPP) modules, insufficient linear aggregation approaches restricting neuronal adaptation, and channel attention mechanisms that inadequately incorporate spatial information—constraints that collectively hamper their ability to fully adapt to complex building structures in remote sensing imagery.

To address these limitations, we propose the dual-attention selective kernel network (DASK-Net), a novel lightweight architecture tailored for remote sensing image segmentation. DASK-Net incorporates two key innovations that directly tackle the shortcomings of existing approaches: First, our efficient inception (EI) unit extracts multiscale features through a computationally efficient design that simulates multiple kernel sizes (1×1, 3×3, 5×5, and 7×7) while minimizing parameter count. This multiscale approach specifically addresses the challenge of capturing objects at different scales in aerial imagery without the computational burden of multiple parallel pathways. Second, our dual-attention selective kernel (DASK) module integrates channel and spatial attention mechanisms in a sequential manner, enabling the network to adaptively focus on both global context and locally salient features. This sequential approach creates a more effective feature refinement process specifically tailored to the challenges of remote sensing imagery, where both spatial details and channel relationships are crucial for accurate segmentation. This architecture enables efficient multiscale feature extraction and refinement while maintaining low computational cost, making it suitable for deployment on resource-constrained platforms such as UAVs.

Comprehensive evaluations, depicted in Fig. 1, demonstrate that our proposed architecture achieves superior performance in terms of IoU, parameter efficiency, and computational cost [giga multiply-accumulate operations (GMACs)] compared to existing lightweight semantic segmentation models on the WHU building test set.

The main contributions of this study are summarized as follows.

1) *EI Unit:* We propose a novel EI unit to extract features across multiple kernels. This approach enables efficient multiscale feature extraction, enhancing the model's ability to capture diverse patterns and structures within the imagery.
2) *Dual-Attention Selective Kernel Module:* We introduce a DASK module that integrates the EI unit with a cascaded dual-attention mechanism to adaptively focus on both global context and locally salient features. This enhances feature extraction by focusing on the prominent features in the channels and spatial locations.
3) *Integration into DASK-Net:* We incorporate the EI unit and the DASK module into a U-shaped framework, resulting in the novel lightweight DASK-Net model. This model generates high-quality segmentation maps while maintaining low computational cost, making it suitable for resource-constrained devices.
4) *Experimental validation:* We conducted extensive experiments on three benchmark datasets to validate the performance of DASK-Net against numerous state-of-the-art methods. The results demonstrate that DASK-Net performs substantially better than other lightweight methods and achieves competitive results compared to more sophisticated networks.

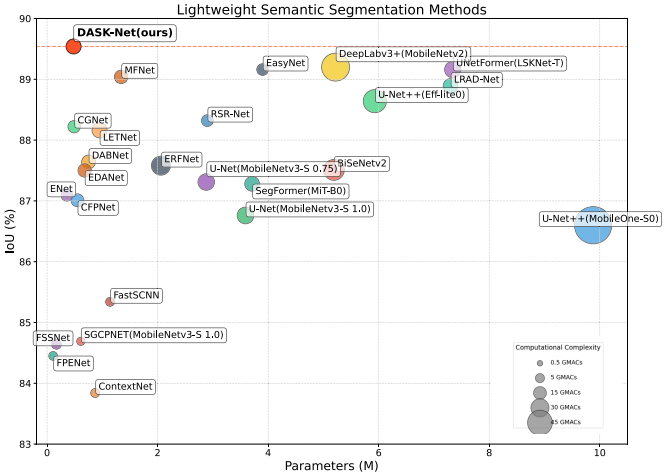


Fig. 1. Comparative evaluation of lightweight semantic segmentation models across IoU, parameter count, and GMACs on the WHU building test set, with an image resolution of 512×512.

The rest of this article is structured as follows. Section II reviews related work, providing context and background. Section III outlines the materials and methods, detailing the proposed DASK-Net. Section IV presents and discusses the experimental results. Finally, Section V offers a brief conclusion.

II. Related Works

Semantic segmentation is a crucial computer vision task that involves classifying each pixel in an image into predefined categories, effectively partitioning the image into meaningful regions. This task belongs to the broader family of dense prediction problems, where predictions are made for every pixel in the input image. Semantic segmentation is also commonly referred to as dense classification, dense prediction, or pixel-wise dense prediction. In this article, we will use these terms interchangeably to refer specifically to the semantic segmentation task.

A. Road Network Extraction From Remote Sensing Imagery

In DL-based semantic segmentation for road extraction from remote sensing imagery, encoder-decoder architectures have gained widespread adoption due to its superior feature extraction capabilities. D-LinkNet [20], winner of

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the 2018 DeepGlobe Road Extraction Challenge [21], employs a pre-trained residual networks with 34 layers (ResNet-34) backbone with dilated convolution layers to expand the receptive field. However, these dilated convolutions can compromise pixel continuity and introduce irrelevant context, affecting road connectivity.

Researchers have proposed various enhancements to address these limitations. Gong et al. [22] implemented dense dilated convolutions with a visual geometry group (VGG) encoder and channel attention to handle occlusions and scale variations. Feature-enhanced dilated LinkNet (FE-LinkNet) [23] improved D-LinkNet through dense connections in downsampling and restructured D-blocks with global pooling. SDG-DenseNet [24] introduced a novel architecture for low-grade road extraction and pioneered decision-level fusion of optical and synthetic aperture radar (SAR) imagery to detect obscured roads.

More sophisticated models have emerged to further enhance performance. Location guided network (LGNet) [25] incorporates an auxiliary road location prediction task to capture global connectivity efficiently. SPIN Road Mapper [26] adds graph reasoning modules to extract long-range dependencies, while cascaded attention dense U-Net (CADUNet) [27] and occlusion-aware road extraction network (OARENet) [28] integrate attention mechanisms to recover occluded road areas. Road augmented deformable attention network (RADANet) [29] employs strip convolutions with deformable attention to capture road shape and pixel dependencies.

For computational efficiency, lightweight road detection network (LRDNet) [30] introduces a multiscale convolutional attention network with a coupled decoder for simultaneous surface, edge, and centerline detection. Lightweight global road feature extraction network (LGRF-Net) [31] utilizes a light ghost basic block with multiscale feature fusion and parallel attention mechanisms, efficiently capturing road structure while significantly reducing model parameters.

B. Building Extraction From Remote Sensing Imagery

Buildings in remote sensing imagery exhibit diverse dimensions and geometries, posing major challenges for automated detection systems that multiscale approaches address by analyzing different spatial resolutions simultaneously. Yuan [32] introduced a deep Lightweight global road feature extraction network (CNN) integrating multi layer activations with a signed distance function for robust boundary preservation. MAP-Net [1] proposed a multiparallel path architecture with fixed-resolution branches, refined by channel-wise attention and pyramid spatial pooling to enhance segmentation across scales. Zhang et al. [33] developed a dual-stream network for capturing both local details and long-range context, while Wu et al. [34] employed a topography-aware multi resolution fusion strategy with a specialized loss for boundary preservation. Multiscale receptive field network (MSRF-Net) [35] used a multiscale receptive field encoder and multipath decoder to recover spatial details effectively.

Attention mechanisms further enhanced performance. Liu et al. [36] designed a multiregion scale-aware network that employs an attention module with a graph-based structure to model inter-scale interactions. In parallel, Gong et al. [37] proposed a context–content collaborative network that integrates context- and content-aware blocks with an edge residual refinement module for balanced precision and completeness. Dai et al. [38] addressed noise and feature loss by using a multiscale location attention network with a location channel attention unit and dual-branch aggregation, while uncertainty-aware network (UANet) [39] tackled uncertainty by fusing multiscale features through a prior information guide module and an uncertainty ranking algorithm.

Hybrid architectures such as BuildFormer [15] introduced a ViT with a dual-path structure and window-based multihead self-attention for accurate building extraction. Dual spatial attention transformer network (DSAT-Net) [40] merged CNNs and transformers via a dual-spatial attention mechanism, fusing global and local paths through a channel mixing refinement module. Bi-branch cross-fusion transformer network (BCT-Net) [41] employed parallel convolutional and transformer branches, merging outputs with an adaptive gating module for improved footprint extraction.

While RSR-Net [17], LRAD-Net [19], and easy network (Easy-Net) [42] incorporate mechanisms like squeeze-and-excitation (SE) blocks, ASPP, and simplified fusion strategies, they struggle with adaptive receptive fields, spatial relationships, and effective multiscale processing. DASK-Net addresses these challenges with an EI unit that extracts multiscale features efficiently and a DASK module that integrates both channel and spatial attention while adaptively adjusting receptive fields. This design enhances feature representation across diverse building types while maintaining a lightweight architecture with significantly fewer parameters.

C. Attention Mechanisms and Multiscale Architectures

Attention mechanisms and multiscale feature extraction significantly impact semantic segmentation for remote sensing imagery. Channel attention, pioneered by squeeze-and-excitation networks [43], recalibrates feature channel importance through global pooling and channel-wise gating. Spatial attention mechanisms like global context network (GCNet) [44] generate masks highlighting salient regions, while combined approaches such as convolutional bottleneck attention module (CBAM) [45] apply both attention types sequentially. However, CBAM applies attention to only a single feature set, limiting its effectiveness when handling complex scenes with varying object characteristics. Although transformer-based self-attention mechanisms [16] excel at capturing global context, their computational demands often restrict deployment in resource-constrained environments.

Multiscale feature extraction addresses scale variations in objects through various approaches. Inception modules [46] introduced parallel convolutional paths with multiple kernel sizes, later evolved in approaches like ASPP in DeepLabv3 [47], which can introduce computational overhead that restricts deployment in resource-constrained environments. Adaptive receptive field approaches like SK networks [48] dynamically adjust feature extraction based on input content, though they rely solely on channel attention for kernel selection, potentially missing critical spatial context. Split-attention network based on ResNet (residual network) architecture (ResNeSt) [49] and sample consistency network (SCNet) [50] refine these concepts through feature partitioning and integrated spatial attention.

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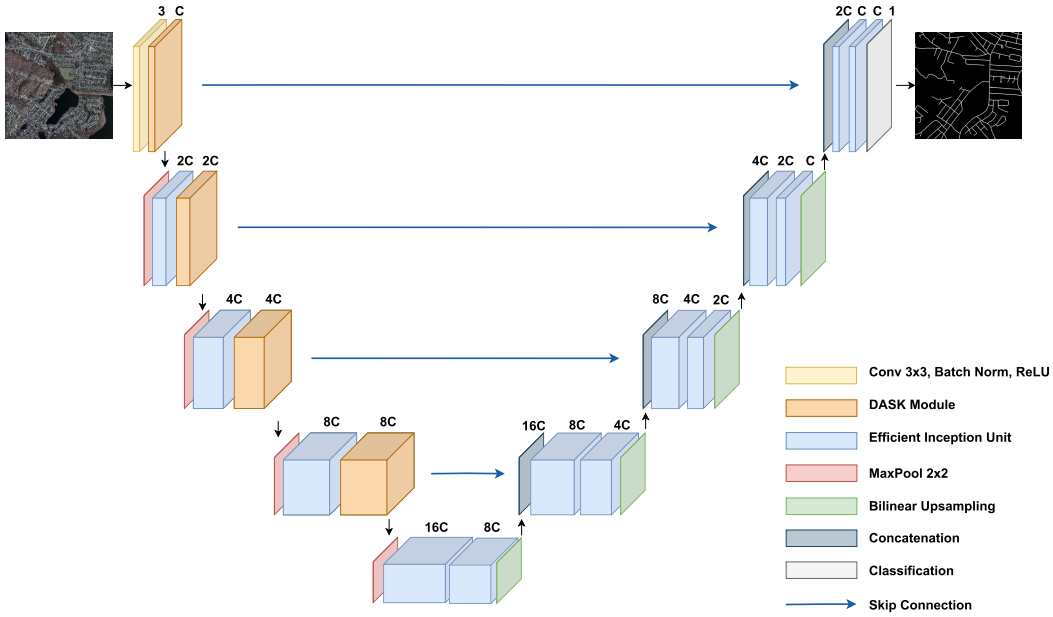


Fig. 2. Illustration of the overall structure of the DASK-Net.

Our DASK-Net addresses these limitations through two key innovations. First, the EI unit provides a parameter-efficient approach to multiscale feature extraction by simulating four distinct kernel sizes (1×1 through 7×7) using sequential convolutions rather than parallel paths. This design reduces parameters compared to traditional inception modules while capturing diverse spatial patterns.

Second, the DASK module integrates spatial and channel attention with adaptive receptive field selection. Unlike SKNet, which uses only channel attention for kernel selection, potentially missing spatial context, and unlike CBAM, which applies attention to a single feature set, our DASK module applies attention mechanisms to features extracted at multiple scales. This enables simultaneous optimization of channel importance, spatial relevance, and receptive field size for each image region—critical for remote sensing imagery where object characteristics vary substantially across scenes.

The DASK module incorporates strategic residual connections within both attention mechanisms and between the input and output paths, improving gradient flow during training and enabling faster convergence. These architectural choices position DASK-Net as an optimal solution for resource-constrained remote sensing applications, achieving superior segmentation quality with minimal computational overhead.

D. Multimodal Semantic Segmentation Methods

Fusion of optical images with modalities such as radar, LiDAR, and thermal has proven to be effective for remote sensing image analysis. Radar imagery, unaffected by lighting and weather, provides robust geometric information despite limited textural detail, making its fusion with optical data valuable for scene understanding.

Hybrid attention-aware fusion network (HAFNet) [51] fused light detection and ranging (LiDAR) and red, green, blue (RGB) using a hybrid attention-aware fusion network with three parallel branches, leveraging an attention-aware fusion block to enhance complementary feature selection. Multitask aware network (MTANet) [52] used hierarchical multimodal fusion and multitask supervision for urban scene understanding. Cross-layer interaction and multiscale fusion network (CIMFNet) [53] explored RGB-digital surface model (DSM) feature interactions through hierarchical interactions and a two-way pyramid pooling module to recover spatial details. Modal memory sharing and morphological complementary networks (MMSMCNets) [54] introduced a modal memory sharing module and a decoding union unit to reinforce morphological features with a novel supervision strategy.

Collectively, these studies demonstrate that while integrating diverse data sources enriches building feature representation and improves extraction accuracy, it also introduces increased computational complexity and potential conflicts among features. Balancing these trade-offs is essential for advancing robust, multimodal building extraction methods.

III. Methodology

This section introduces the proposed DASK-Net architecture, highlighting its innovative components. We begin with a high-level overview of the network and detailed explanations of its two key elements: the EI unit and the double attention SK module.

A. Overview

We propose the DASK-Net architecture to improve a network's efficiency in capturing feature maps from various receptive fields while enhancing these maps channel-wise and spatial-wise. As illustrated in Fig. 2, our approach employs a U-shaped framework consisting of four encoders and four decoder stages, interconnected by skip connections and a bridging layer at the bottom stage.

The DASK-Net architecture utilizes a symmetric encoder–decoder structure for image segmentation. The

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TABLE I
DETAILED ARCHITECTURAL CONFIGURATION OF DASK-NET

ENCODER				DECODER			
Stage	**Operation**	**Channel**	**Output**	**Stage**	**Operation**	**Channel**	**Output**
Input			512x512x3	Classification	Conv 1x1	32 → 1	512x512x1
E 11	Conv 3x3	3 → 32	512x512x32	D 12	EI	32 → 32	512x512x32
E 12	DASK	32 → 32	512x512x32	D 11	Concat + EI	64 → 32	512x512x32
Downsampling	MaxPool 2x2		256x256x32	Upsampling	BI		512x512x32
E 21	EI	32 → 64	256x256x64	D 22	EI	64 → 32	256x256x32
E 22	DASK	64 → 64	256x256x64	D 21	Concat + EI	128 → 64	256x256x64
Downsampling	MaxPool 2x2		128x128x64	Upsampling	BI		256x256x64
E 31	EI	64 → 128	128x128x128	D 32	EI	128 → 64	128x128x64
E 32	DASK	128 → 128	128x128x128	D 31	Concat + EI	256 → 128	128x128x128
Downsampling	MaxPool 2x2		64x64x128	Upsampling	BI		128x128x128
E 41	EI	128 → 256	64x64x256	D 42	EI	256 → 128	64x64x128
E 42	DASK	256 → 256	64x64x256	D 41	Concat + EI	512 → 256	64x64x256
Downsampling	MaxPool 2x2		32x32x256	Upsampling	BI		64x64x256
Bridge 1	EI	256→ 512	32x32x512	Bridge 2	EI	512 → 256	32x32x256

encoder extracts spatial features, while the decoder reconstructs the segmentation map. The encoder's first stage has two units: a conventional 3×3 convolution with rectified linear unit (ReLU) activation and *batch normalization*, followed by a novel DASK module. Subsequent encoder stages consistently apply downsampling and feature extraction. Each stage begins with a max-pooling operation that halves the spatial dimensions of the input feature map. This downsampled input is then processed sequentially through an EI unit, followed by a DASK module. At each encoder stage, the number of channels doubles, while the height and width of the feature maps are halved. Outputs from each encoder stage connect to the corresponding decoder stages via skip connections. It is important to note that the DASK units are strategically integrated into the latter part of each decoder stage, enhancing the network's ability to highlight salient features transmitted via skip connections to the corresponding decoder stage. This placement ensures that the information from the encoder undergoes adaptive feature refinement before being utilized in the final segmentation process.

A bridge module links the encoder and decoder, consisting of a max-pooling layer, two subsequent EI units, and a bilinear upsampling operation. In the decoder, each stage's input is concatenated with the corresponding encoder output through skip connections and then processed by subsequent EI units and a bilinear upsampling operation. At each decoder stage, the channel depth halves, while spatial resolution doubles. The final stage includes an additional 1×1 convolution to generate the final segmentation map. This structure enables DASK-Net to effectively capture adaptive multiscale features, balancing deep feature representation with detailed spatial information, which is crucial for accurate image segmentation.

Table I highlights the detailed structure and layer-wise implementation of the DASK-Net model, where E and D denote encoder and decoder layers, while the first and second numbers following them represent the stage and layer ordering numbers, respectively.

B. EI Unit

To address the unique challenges of multiscale feature extraction in remote sensing imagery, we propose the EI unit, a novel architectural component designed to capture diverse spatial contexts while optimizing computational efficiency. Unlike the naive inception module [46], our design leverages sequential convolutions to simulate larger kernels, processing input through two parallel paths: a 1×1 convolution and a simulated 7×7 convolution. As highlighted in Fig. 3, given an input feature map $X \in \mathbb{R}^{H \times W \times C}$ and output feature map $Y \in \mathbb{R}^{H \times W \times K}$, the EI Unit processes X as follows.

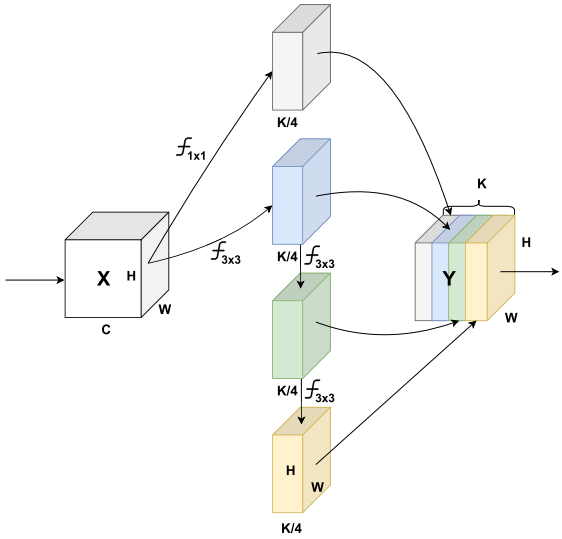


Fig. 3. Illustration of the proposed EI unit.

Path 1 (1×1 Convolution):

$$Z_{1\times1} = \text{Conv}_{1\times1}(X) \in \mathbb{R}^{H\times W\times K/4}. \tag{1}$$

Paths 2 (7×7 Convolution Through Sequential ConvMixer Operations):

$$Z_{3\times3} = \text{ConvMixer}(X) \in \mathbb{R}^{H\times W\times K/4} \tag{2}$$

$$Z_{5\times5} = \text{ConvMixer}(Z_{3\times3}) \in \mathbb{R}^{H\times W\times K/4} \tag{3}$$

$$Z_{7\times7} = \text{ConvMixer}(Z_{5\times5}) \in \mathbb{R}^{H\times W\times K/4}. \tag{4}$$

Each ConvMixer block implements

$$\text{ConvMixer}(Z) = \text{ReLU}\big(\text{BN}(\text{PWConv}_{1\times1}(Z_{\text{res}}))\big)$$
$$\text{where} \quad Z_{\text{res}} = \text{ReLU}\big(\text{BN}(\text{DWConv}_{3\times3}(Z))\big) + Z \tag{5}$$

where $\text{DWConv}_{3\times3}$ is a depthwise convolution with 3×3 kernel, $\text{PWConv}_{1\times1}$ is a pointwise (1×1) convolution, BN is batch

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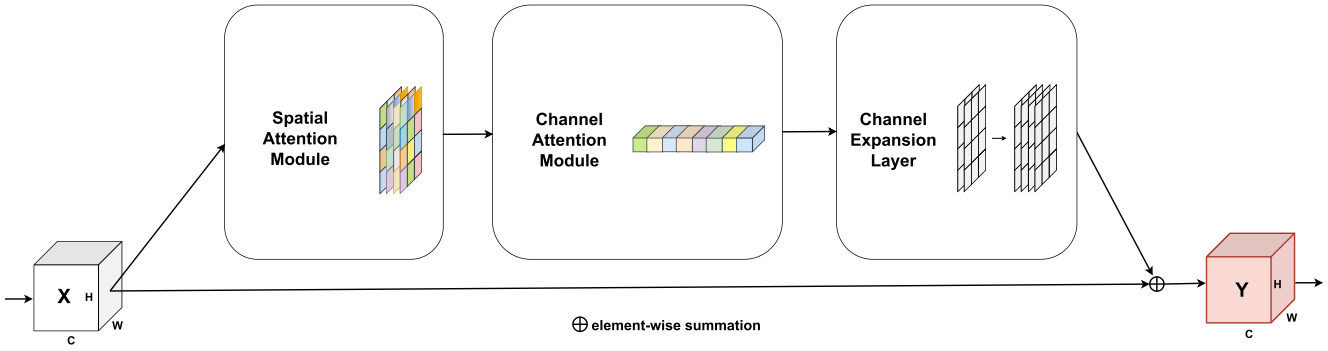


Fig. 4. Illustration of the proposed double attention SK module.

normalization, ReLU is the rectified linear unit activation, and Z_{res} represents the output of the residual connection that adds the input to the processed feature maps.

The final output concatenates all paths

$$Y = \text{Concat}(Z_{1\times1}, Z_{3\times3}, Z_{5\times5}, Z_{7\times7}) \in \mathbb{R}^{H\times W\times K}. \quad (6)$$

This design effectively simulates receptive fields of sizes 1×1, 3×3, 5×5, and 7×7 through the sequential application of 3×3 ConvMixer blocks, capturing multiscale features while maintaining computational efficiency. The use of depthwise separable convolutions and residual connections further reduces parameters and improves gradient flow during training.

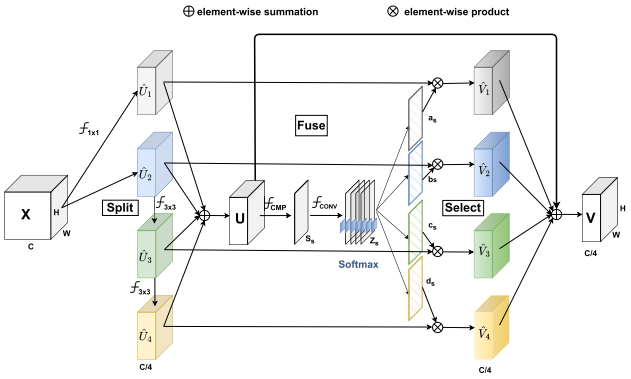


Fig. 5. Illustration of the proposed SAM.

C. Double Attention Selective Kernel Module

Building upon the work of Li et al. [48], who introduced SK convolution to address the issue of fixed receptive fields at varying input scales, we propose the DASK module. As illustrated in Fig. 4, this module enhances the adaptive capabilities of SK convolution by extending its original channel-only attention mechanism with an additional spatial attention module (SAM).

In semantic segmentation, the task encompasses two critical aspects: object classification and spatial localization. The former involves identifying the semantic categories present in the image, while the latter requires precise delineation of object boundaries at the pixel level. The channel attention module (CAM) in DASK selectively emphasizes informative feature channels by learning channel-wise importance weights, enabling the network to focus on the most discriminative features for object recognition. Concurrently, the SAM generates a pixel-wise attention map, highlighting salient regions within each feature map. This spatial weighting allows the network to concentrate on areas of high semantic relevance, potentially corresponding to object boundaries or distinctive textures. Combining these two attention mechanisms enables the DASK module to adaptively refine feature representations in both the channel and spatial dimensions, potentially improving the network's ability to capture fine-grained object details while maintaining sensitivity to the global context.

Furthermore, we integrate the EI unit into a multibranch module, enhancing the model's ability to process inputs efficiently across various pathways. The DASK module can replace conventional convolutions in existing architectures with minimal computational overhead, improving adaptability to varying input scales and spatial distributions of features.

1) Spatial Attention Module: Spatial attention in semantic segmentation enhances model performance by enabling it to focus on the most relevant image regions, thereby improving accuracy and efficiency in handling complex scenes. This technique is crucial for achieving fine-grained segmentation results, managing occlusions, and providing interpretable insights into the model's decision-making process. Fig. 5 illustrates the proposed SAM in the novel DASK-Net architecture. To generate a spatially enhanced feature map, the input undergoes three critical operations: *Split*, *Fuse*, and *Select*.

Split: Given input $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent height, width, and channel, respectively, we apply the EI unit to obtain four transformations $\hat{U}_n \in \mathbb{R}^{H\times W\times C/4}$ as follows:

$$\hat{U}_1 = \text{ReLU}\left(\text{BN}\left(f_{1\times1}^{C\mapsto C/4}(X)\right)\right) \quad (7)$$

$$\hat{U}_2 = f_{3\times3}^{C\mapsto C/4}(X) \quad (8)$$

$$\hat{U}_3 = f_{3\times3}^{C/4\mapsto C/4}(\hat{U}_2) \quad (9)$$

$$\hat{U}_4 = f_{3\times3}^{C/4\mapsto C/4}(\hat{U}_3) \quad (10)$$

where $f_{1\times1}$ is a 1×1 convolution operation followed by a BN layer and ReLU activation function, while $f_{3\times3}$ is a ConvMixer layer with a 3×3 kernel size. The decomposed individual features are stacked to form a list of transformation

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maps from varying receptive fields as follows:

$$\hat{U} = \text{Stack}\big[\hat{U}_1, \hat{U}_2, \hat{U}_3, \hat{U}_4\big]. \tag{11}$$

Fuse: Next, we perform an element-wise summation operation on the stacked $\hat{U}$ to obtain a feature map that contains information from all branches

$$U = \sum_{i=1}^{4} \hat{U}_i. \tag{12}$$

The fused feature map then undergoes channel-wise max pooling (f_{CMP}) to extract distinctive object features $S \in \mathbb{R}^{H\times W\times 1}$, enabling more refined spatial attention inference. The equation to obtain the ijth element of S can be expressed as

$$S_{ij} = \max\{U_{ij}^{k} : k = 1, \ldots, c\} \tag{13}$$

where c is the channel size of the fused feature map U. The resultant feature map contains the most salient features from all four branches. Next, we perform a 1×1 convolution operation to obtain $Z \in \mathbb{R}^{H\times W\times 4}$ which is used to compute the inter-spatial dependencies among branches. The equation is expressed as follows:

$$Z = f_{1\times 1}^{1\mapsto 4}(S). \tag{14}$$

Furthermore, we apply the Softmax operation along the channel dimension of Z, followed by a split operation to retrieve an individual spatial selection mask for each decomposed branch

$$\hat{Z} = \sigma(Z) \tag{15}$$

$$\hat{Z} = \text{split}\big(\hat{Z}, [4, 1]\big) \longmapsto [a, b, c, d] \tag{16}$$

where σ denotes the Softmax layer, and the split operation divides the 4-D output into four 1-D vectors $[a, b, c, d] \in \mathbb{R}^{H\times W\times 1}$.

Select: The features from the four branches are weighted by their respective spatial selection masks and then fused through summation to obtain the spatial attention map. For faster convergence and smoother gradient flow, we perform element-wise addition between the fused feature map of decomposed branches U and the obtained attention map. The mathematical expression for obtaining the feature map $V \in \mathbb{R}^{H\times W\times C/4}$ is as follows:

$$V = \sum_{i=1}^{4} \big(\hat{Z}_i \otimes \hat{U}_i\big) + U. \tag{17}$$

2) Channel Attention Module: The CAM dynamically adjusts feature responses across channels by explicitly modeling inter-channel relationships, enabling adaptive feature recalibration. As illustrated in Fig. 6, unlike SAM, CAM inherits the final four masked branch outputs ($\hat{V}_1$, $\hat{V}_2$, $\hat{V}_3$, $\hat{V}_4$) and their fused map V from SAM in the *Split* stage.

Fuse: First, the individual spatially masked features inherited from SAM are stacked together to prepare for the application of channel-wise masks in the *Select* stage

$$\hat{V} = \text{Stack}\big[\hat{V}_1, \hat{V}_2, \hat{V}_3, \hat{V}_4\big] \tag{18}$$

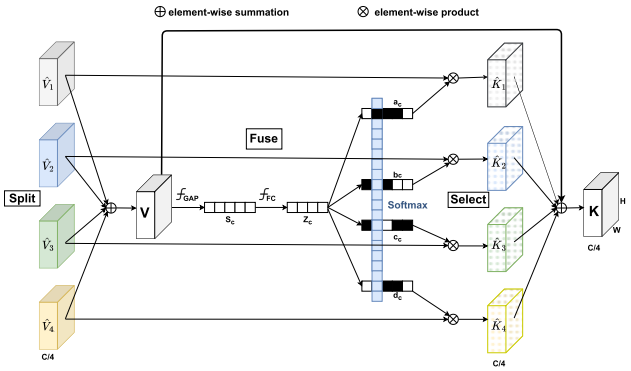


Fig. 6. Illustration of the proposed CAM.

then, the global average pooling operation (f_{GAP}) is performed on V across the channel axis to generate channel-wise statistics

$$S = f_{\text{GAP}}^{H\times W\times C/4\mapsto 1\times 1\times C/4}(V). \tag{19}$$

Next, we apply a fully connected layer, followed by BN and ReLU activation function operations, to create a compact feature map $Z \in \mathbb{R}^{1\times 1\times d}$

$$Z = \text{ReLU}\Big(\text{BN}\Big(f_{\text{FC}}^{C/4\mapsto d}(S)\Big)\Big). \tag{20}$$

Different from SK [48], which sets $d = \max(C/r, 32)$ with a reduction ratio $r = 16$, we set $d = 32$ as C/r never exceeds 32 in our design.

Next, we apply another fully connected layer to the compact feature Z, mapping from d dimensions to C, followed by a Softmax operation along the channel axis. Subsequently, we split the output to obtain a stacked list of individual channel-wise attention masks for each decomposed branch

$$\hat{Z} = \sigma\big(f_{\text{FC}}^{d\mapsto C}(Z)\big) \tag{21}$$

$$\hat{Z} = \text{split}(\hat{Z}, [4, C/4]) \longmapsto [a, b, c, d] \tag{22}$$

where σ denotes the Softmax layer, and the split operation divides the C-dimensional output into four $C/4$-dimensional vectors $[a, b, c, d] \in \mathbb{R}^{1\times 1\times C/4}$.

Select: Furthermore, we apply the obtained channel-wise attention masks to the spatially enhanced feature maps, fuse the results, and perform an element-wise summation with the input V to generate the final output of the CAM

$$K = \sum_{i=1}^{4} \big(\hat{Z}_i \otimes \hat{V}_i\big) + V. \tag{23}$$

3) Channel Expansion Layer: After the input is processed sequentially with SAM and CAM, the resultant feature map K, enriched with spatial and channel adaptive kernels, is further processed with a 1×1 convolution layer. This layer expands the channel size from $C/4$ to the original channel size C of the input X. Finally, we fuse the expanded feature map with the input X through an element-wise summation operation to attain the final output $Y \in \mathbb{R}^{H\times W\times C}$

$$Y = f_{1\times 1}^{C/4\mapsto C}(K) + X. \tag{24}$$

CAM and SAM can be used independently or together, yielding a refined feature map enriched both in channel

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and spatial dimensions. Our ablation study explores various configurations of integrating spatial and channel attention within DASK-Net. We systematically evaluate different orderings and combinations of these mechanisms to assess their individual and combined effects on the model's performance.

IV. Experimental Settings and Results

To rigorously evaluate the efficiency and effectiveness of the proposed DASK-Net model for pixel-wise dense classification tasks in remote sensing imagery, we conducted comprehensive experiments focusing on two critical applications: road extraction and building footprint delineation. We used the Massachusetts roads [75] and DeepGlobe [21] dataset for the road extraction task, and the WHU building dataset [13] for building footprint extraction.

A. Experimental Settings

1) Datasets and Preprocessing: For the road extraction task, we utilized the Massachusetts roads dataset [75] and DeepGlobe road extraction dataset. Massachusetts roads dataset is a comprehensive benchmark consisting of 1171 high-resolution aerial images covering over 2600 km^2 of diverse urban, suburban, and rural landscapes across Massachusetts, USA. The dataset is divided into 1108 training, 14 validation, and 49 test images, each with a resolution of 1500×1500 pixels. Our preprocessing pipeline involved padding the training and validation images to 1536×1536 pixels and partitioning them into nine 512×512 non-overlapping patches, resulting in 9972 training and 126 validation patches. This method ensures a diverse and robust training dataset. For the test set, we applied the same padding technique to generate 1536×1536 pixel images. We calculated the IoU scores on these padded test images to maintain evaluation integrity, excluding the padded regions to prevent artificially inflated performance metrics.

The DeepGlobe dataset [21] comprises 8570 satellite images of resolution 1024×1024 pixels, covering Thailand, Indonesia, and India at 50 cm/pixel resolution. The dataset represents real-world challenges through its diverse geographic coverage, including both rural and urban regions with varying road surfaces (paved, unpaved, and dirt roads). The original split includes 6226 images (1632 km^2) for training, 1243 images (362 km^2) for validation, and 1101 images (288 km^2) for testing. For our experiments, we used only the training set with labeled masks, applying an 80:10:10 split for training, validation, and test sets. Images were divided into non-overlapping 512×512 patches, with roadless patches removed. Our final dataset contains 16 320 images for training, 2016 for validation, and 2039 for testing.

We used the Wuhan (WHU) building dataset [13], which includes satellite and aerial images for the building extraction task. Our study focused exclusively on the aerial image dataset, featuring a high spatial resolution of 0.3 m and encompassing over 22 000 buildings across an area exceeding 450 km^2. This dataset consists of 8189 aerial images, each with a resolution of 512×512 pixels. The dataset is divided into 4736 training, 1036 validation, and 2416 testing images for comprehensive model evaluation.

To improve model generalization, we applied data augmentation techniques to all datasets. Specifically, we randomly applied one of three transformations—horizontal flip, vertical flip, or 90° rotation to each training sample. This strategy introduces variability in the training data, helping the model learn to handle different orientations. Subsequently, we normalized all images using the mean and standard deviation derived from the training set. This normalization ensures consistent input scaling, facilitating more stable and efficient model training.

2) Benchmark Methods: To evaluate DASK-Net's effectiveness, we conducted comprehensive experiments comparing our approach against state-of-the-art semantic segmentation methods. Our evaluation framework addresses two key dimensions: segmentation quality and computational efficiency, reflecting our goal of maintaining high accuracy while significantly reducing model complexity.

The first comparison category encompasses established semantic segmentation architectures renowned for their exceptional performance across diverse domains, alongside recently introduced methodologies specifically designed to address the unique challenges of remote sensing image analysis. These include "U" shaped network (U-Net) [57] with and without lightweight mobile network version 3 (MobileNetV3) backbones, DeepLabv3 [47] with its effective atrous convolution-based multiscale context extraction capabilities, and several recent remote sensing-specific architectures such as separable graph convolutional network (SGCNNet), multiscale and multidirection feature fusion network (MSMDFFNet), and boundary optimization and multiscale context network (BOMSC-Net) that have demonstrated notable performance in aerial and satellite imagery analysis.

The second category consists of lightweight models specifically engineered for resource-constrained scenarios. This category is further stratified based on parameter count: 1) medium-scale models (1–10 million parameters) such as U-Net++ [59], "U" shaped transformer network (UNetFormer) [60], EasyNet [42], LRAD-NET [19], DeepLabv3+ [64], and bilateral segmentation network version 2 (BiSeNetv2) [77] and 2) small-scale models (less than 1 million parameters) such as lightweight real-time semantic segmentation network (LETNet) [78], context network (ContextNet) [67], DABNet [68], efficient dense modules with asymmetric convolution network (EDANet) [69], context guided network (CGNet) [71], spatial-detail guided context propagation network (SGCPNet) [76], channel-wise feature pyramid (CFPNet) [70], and efficient neural network (ENet) [72]. This carefully selected benchmark suite enables a holistic assessment of DASK-Net, revealing its effectiveness in addressing the unique challenges of lightweight semantic segmentation for remote sensing imagery.

3) Implementation Details and Loss Function: In our experimental setup, we utilized PyTorch as the DL framework and conducted all training procedures on a cluster of eight NVIDIA RTX A5000 GPUs. To ensure a fair comparison with baseline approaches, we initialized all models using PyTorch's default settings for all datasets.

We employed a batch size of 32 and an initial learning rate of 0.003 for the Massachusetts roads dataset. We used the Adam optimizer and a cosine learning rate scheduler to

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TABLE II
QUANTITATIVE RESULTS ON THE MASSACHUSETTS ROADS DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
SGCNNet [56]	ResNet50	42.73	170.92	30.05	63.96	79.86	76.26	78.02
DeepLabv3 [48]	ResNet50	39.64	151.43	164.14	63.02	76.63	78.02	77.32
MSMDFFNet [57]	ResNet34	39.26	149.88	149.602	**66.16**	**80.71**	**78.59**	**79.64**
U-Net [58]	-	31.04	118.04	218.95	65.87	80.49	78.38	79.42
LinkNet [59]	ResNet18	**11.53**	**44.04**	**12.08**	64.38	80.26	76.5	78.33
medium								
U-Net++ [60]	MobileOne-S0	9.88	38.06	45.36	65.4	**80.9**	77.34	79.08
UNetFormer [61]	LSKNet-T	7.34	28.58	9.43	63.78	79.03	76.77	77.88
U-Net++ [60]	EfficientNet-lite0	5.93	22.8	22.19	65.2	80.08	77.82	78.94
MACUNet [62]	-	5.15	19.67	33.59	**65.44**	79.79	**78.45**	**79.11**
SegFormer [63]	MiT-B0	3.71	14.17	6.74	60.79	75.22	76.02	75.62
U-Net [58]	MobileNetV3-Small 1.0	3.59	13.73	10.22	63.14	78.43	76.41	77.41
U-Net [58]	MobileNetV3-Small 0.75	2.88	11.04	9.82	63.79	78.85	76.95	77.89
ERFNet [64]	-	2.06	7.9	14.73	65.01	79.82	77.8	78.79
DeepLabV3+ [65]	MobileNetV3-Small 0.75	1.62	6.23	2.62	57.2	73.47	72.09	72.78
MFNET [66]	-	1.34	5.12	4.60	64.33	79.4	77.22	78.29
FastSCNN [67]	-	**1.14**	**4.42**	**0.88**	52.33	68.62	68.8	68.71
small								
ContextNet [68]	-	0.87	3.42	0.89	53.11	67.72	71.13	69.38
DABNet [69]	-	0.75	2.9	5.28	63.88	76.57	**79.41**	77.96
EDANet [70]	-	0.68	2.61	4.45	55.84	70.7	72.64	71.66
SGCPNET [56]	MobileNetV3-Small 1.0	0.61	2.37	0.44	59.79	73.37	76.36	74.84
CFPNet [71]	-	0.55	2.12	4.00	62.95	76.05	78.52	77.26
CGNet [72]	-	0.49	1.92	3.55	63.67	76.44	79.21	77.8
ENet [73]	-	0.36	1.39	2.84	64.69	79.8	77.36	78.56
FSSNet [74]	-	0.17	0.68	1.37	64.22	78.66	77.77	78.21
FPENet [75]	-	**0.11**	**0.46**	**0.78**	60.66	76.27	74.76	75.51
DASK-Net(ours)	-	0.48	1.91	4.85	**65.47**	**80.85**	77.49	**79.13**

dynamically adjust the learning rate throughout the training process. All models were trained from scratch for 200 epochs to ensure full convergence. For the WHU building dataset, we adjusted our hyperparameters to suit the dataset's characteristics. We set the batch size to 16 and used an initial learning rate of 0.001 with a weight decay factor 0.0001. The AdamW optimizer was chosen for model optimization, and we employed a dynamic learning rate adjustment strategy, reducing the rate by a factor of 0.1 if the validation loss showed no improvement over ten consecutive epochs. The training was conducted from scratch for 150 epochs, with the model state achieving the lowest validation loss saved as a checkpoint for inference. For the DeepGlobe dataset, our implementation maintained consistency with the WHU dataset configuration, with the sole modification being an increased batch size of 32.

To address the substantial class imbalance inherent in our high-resolution remote sensing datasets, we implemented a strategic loss function approach. While binary cross-entropy (BCE) is typically favored for its stability in balanced binary segmentation tasks, it struggles with highly imbalanced data. Conversely, Dice loss excels in handling class imbalance by evaluating the overlap between predicted and ground truth samples. Acknowledging the complementary benefits of these approaches, we opted for a Combo loss function [79], which effectively integrates BCE and Dice loss to leverage their respective strengths. This hybrid approach combines the stability of BCE with the class imbalance robustness of Dice loss, enabling a more comprehensive training objective. By doing so, we aimed to enhance the model's ability to accurately segment both the minority class (roads or buildings) and the majority class (background), thereby improving overall segmentation performance in our challenging datasets.

4) Evaluation Metrics: To evaluate the performance of our proposed model and the benchmarking methods, we employed four metrics—precision, recall, $F1$ score, and Jaccard index. Precision measures the proportion of accurately predicted positive pixels out of all pixels predicted as positive by the model. At the same time, recall evaluates the proportion of accurately predicted positive pixels out of all pixels that are actually positive. The $F1$ score, the harmonic mean of precision and recall, provides a balanced measure of model performance, mitigating the impact of extreme values in either component and offering a unified metric for scenarios where both high precision and high recall are crucial

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{25}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{26}$$

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{27}$$

$$\text{Jaccard}(A, B) = \frac{|A \cap B|}{|A| \cup |B|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{28}$$

The Jaccard index, also known as IoU, evaluates model accuracy by computing the overlap between the predicted mask and the ground truth, divided by their total combined area, offering a precise measure of spatial alignment.

B. Results on Massachusetts Roads Dataset

The results in Table II demonstrate DASK-Net's strong performance on the Massachusetts roads dataset while maintaining exceptional parameter efficiency. With only 0.48M parameters (1.91 MB model size), DASK-Net achieves competitive accuracy metrics of 65.47% IoU and 79.13% $F1$ score.

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TABLE III
QUANTITATIVE RESULTS ON THE DEEPGLOBE DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
DeepLabv3 [48]	ResNet50	39.64	151.43	164.14	63.02	76.63	78.02	77.32
MSMDFFNet [57]	ResNet34	39.26	149.88	149.602	**67.67**	**81.55**	**79.91**	**80.72**
U-Net [58]	-	31.04	118.04	218.95	65.28	80.92	77.16	78.99
LinkNet [59]	ResNet18	**11.53**	**44.04**	**12.08**	64.04	79.21	76.98	78.08
medium								
U-Net++ [60]	MobileOne-S0	9.88	38.06	45.36	64.92	**79.51**	77.97	78.73
UNetFormer [61]	LSKNet-T	7.34	28.58	9.43	**65.38**	79.21	**78.92**	**79.07**
U-Net++ [60]	EfficientNet-lite0	6.7	25.77	22.80	64.87	78.72	78.67	78.69
SegFormer [63]	MiT-B0	3.71	14.17	6.74	61.83	75.27	77.59	76.41
U-Net [58]	MobileNetV3-Small 1.0	3.59	13.73	10.22	63.4	78.59	76.64	77.6
U-Net [58]	MobileNetV3-Small 0.75	2.88	11.04	9.82	63.53	78.83	76.6	77.7
DeepLabV3+ [65]	MobileNetV3-Small 0.75	1.62	6.23	2.62	61.87	77.53	75.39	76.44
MFNET [66]	-	1.34	5.12	4.60	64.26	78.9	77.59	78.24
FastSCNN [67]	-	**1.14**	**4.42**	**0.88**	61.92	76.9	76.07	76.49
small								
ContextNet [68]	-	0.87	3.42	0.89	59.77	75.88	73.79	74.82
DABNet [69]	-	0.75	2.9	5.28	64.24	78.34	78.11	78.23
EDANet [70]	-	0.68	2.61	4.45	64.62	78.06	**78.97**	78.51
SGCPNET [78]	MobileNetV3-Small 1.0	0.61	2.37	0.44	62.83	77.75	76.6	77.17
CFPNet [71]	-	0.55	2.12	4.00	63.99	77.78	78.31	78.04
CGNet [72]	-	0.49	1.92	3.55	64.63	78.36	78.67	78.51
ENet [73]	-	0.36	1.39	2.84	63.86	77.08	78.82	77.94
FSSNet [74]	-	0.17	0.68	1.37	60.81	75.29	75.98	75.63
FPENet [75]	-	**0.11**	**0.46**	**0.78**	62.02	77.02	76.1	76.56
DASK-Net(ours)	-	0.48	1.91	4.85	**65.54**	**80.07**	78.31	**79.18**

Notably, DASK-Net achieves the second-highest precision (80.85%) among all compared methods, just slightly behind U-Net++ (80.9%).

In the small-model category, DASK-Net significantly outperforms most baselines. While DABNet [68] achieves higher recall (79.41%), DASK-Net delivers better overall performance with superior precision and $F1$ score. Similarly, CGNet [71] shows strong recall (79.21%) but falls short of DASK-Net's precision and $F1$ score.

Compared to larger models, DASK-Net maintains competitive performance despite dramatic reductions in model complexity. U-Net [57] achieves marginally better IoU (65.87%) and $F1$ score (79.42%), but it requires 31.04M parameters—approximately 65 times more than DASK-Net. Similarly, multiscale skip connected and asymmetric-convolution-based U-Net (MACUNet) [61] provides comparable IoU (65.44%) but with over ten times the parameters.

In terms of computational efficiency, DASK-Net requires 4.85 GMACs, representing a reasonable trade-off between performance and computational demands. While models like SGCPNET [76], FPENet [74], and fast semantic segmentation network (FastSCNN) [66] require fewer computations (0.44, 0.78 and 0.88 GMACs, respectively), they deliver substantially lower accuracy metrics. DASK-Net achieves over 10% higher $F1$ score than FastSCNN while maintaining similar parameter efficiency.

Fig. 7 highlights the visual comparison of the selected benchmark models and the proposed DASK-Net on randomly selected images from the test set. The results show that our proposed method outperforms the benchmark methods, generating more accurate and finer segmentation maps due to its adaptive receptive fields and double-attention mechanism.

C. Results on the DeepGlobe Dataset

Table III presents the comprehensive results of our experiments on the DeepGlobe dataset, highlighting the remarkable performance of the proposed DASK-Net model. Despite belonging to the small model category with only 0.48M parameters, DASK-Net demonstrates exceptional effectiveness, achieving an $F1$ score of 79.18% and precision of 80.07%, which rivals or surpasses many larger models.

In the large model category, MSMDFFNet [56] achieves the highest $F1$ score of 80.72% with 39.26M parameters, while LinkNet [58] provides the most parameter-efficient approach with 11.53M parameters. However, DASK-Net delivers comparable performance with just 1.2% of MSMDFFNet's parameter count, demonstrating the efficiency of our architecture design.

Medium-sized models like UNetFormer [60] achieve respectable results with an $F1$ score of 79.07%, but still require significantly more parameters (7.34M) than DASK-Net. FastSCNN [66], though relatively lightweight in this category with 1.14M parameters, achieves a lower $F1$ score of 76.49%.

Among small models, DASK-Net outperforms all baselines with the highest IoU (65.54%), precision (80.07%), and $F1$ score (79.18%). This performance is particularly noteworthy given that DASK-Net maintains a modest computational complexity of just 4.85 GMACs while delivering segmentation accuracy that approaches state-of-the-art large models.

These experimental results demonstrate that the proposed DASK-Net achieves an optimal balance between model efficiency and segmentation performance. By maintaining competitive accuracy metrics with minimal computational requirements, DASK-Net proves highly suitable for resource-constrained deployment scenarios without sacrificing segmentation quality.

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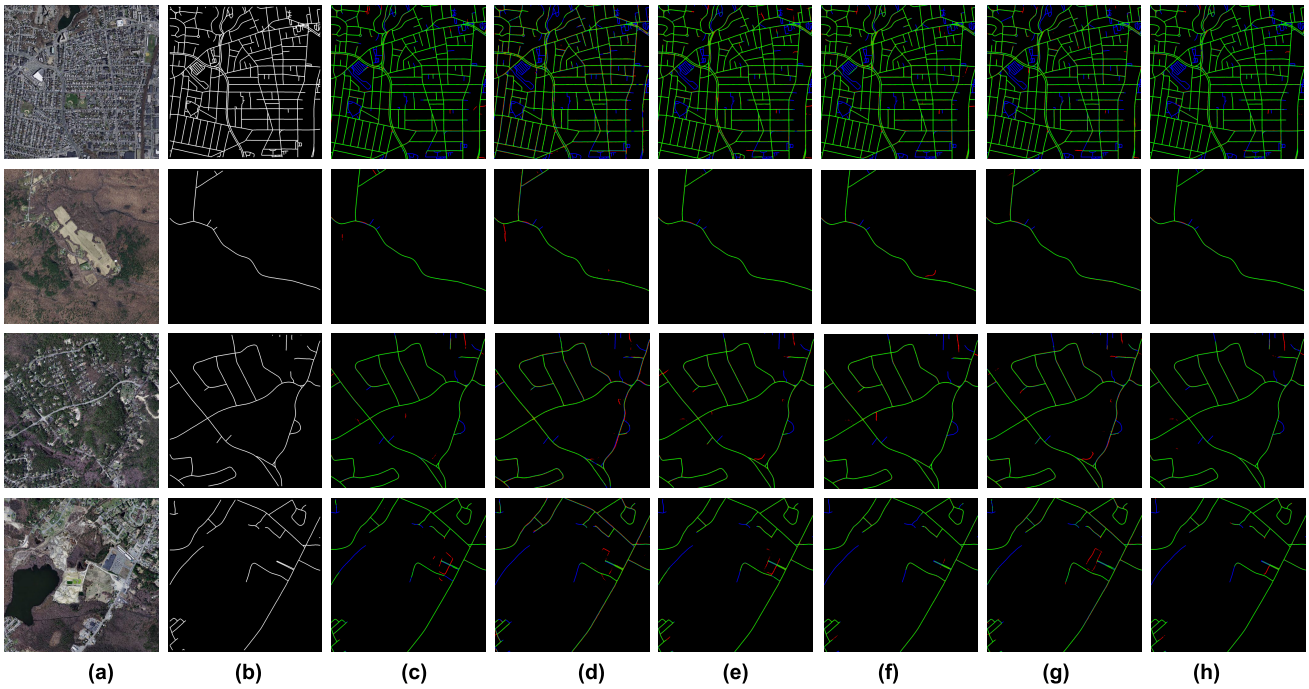


Fig. 7. Visual comparison on the Massachusetts roads dataset. (a) Original image. (b) Original segmentation mask. (c) U-Net. (d) FastSCNN. (e) DABNet. (f) CGNet. (g) FPENet. (h) DASK-Net(ours). False positive pixels are marked in red, false negative pixels are marked in blue, and true positive pixels are marked in green.

TABLE IV
QUANTITATIVE RESULTS ON THE WHU BUILDING DATASET

Methods	Backbone	Params(M)	Size(MB)	MACs(G)	IoU	Precision	Recall	F1 Score
large								
BOMSC-Net* [82]	-	129.32	517.28	-	90.15	95.14	94.5	94.8
LFEMAP-Net* [83]	ConvNeXt-B	127.37	509.48	-	**91.09**	**95.81**	94.86	**95.34**
DSAT-Net [41]	ConvNext-Tiny	48.50	202.12	57.60	86.24	91.11	94.16	92.61
BuildFormer [15]	-	40.52	154.99	116.58	90.58	95.17	**94.95**	95.06
DeepLabv3 [48]	ResNet50	39.64	151.43	164.14	85.69	91.73	92.86	92.29
U-Net [58]	-	31.04	118.04	218.95	87.79	93.02	93.99	93.5
UANet [40]	ResNet50	26.73	102.19	29.81	89.45	94.54	94.32	94.43
LinkNet [59]	ResNet18	**11.53**	**44.04**	**12.08**	84.02	89.99	92.69	91.32
medium								
U-Net++ [60]	MobileOne-S0	9.88	38.06	45.36	86.6	94.07	92.72	93.39
UNetFormer [61]	LSKNet-T	7.34	28.58	9.43	89.16	94.77	93.77	94.27
LRAD-Net* [20]	Reg-Net 600	7.3	29.2	6.68	88.89	94.21	93.74	93.86
U-Net++ [60]	EffecientNet-lite0	5.93	22.8	22.19	88.64	94.03	93.93	93.98
DeepLabv3+ [65]	MobileNetv2	5.22	20.05	30.95	**89.2**	94.19	94.45	**94.29**
BiSeNetv2 [79]	-	5.19	19.87	17.79	87.51	93.43	93.25	93.34
EasyNet* [43]	RegNet08	3.90	15.6	3.00	89.16	93.48	**94.94**	94.2
SegFormer [63]	MiT-B0	3.71	14.17	6.74	87.28	93.43	92.99	93.21
U-Net [58]	MobileNetv3-Small 1.0	3.59	13.73	10.24	86.76	93.06	92.76	92.91
RSR-Net* [18]	-	2.9	11.6	3.01	88.32	**94.92**	92.63	-
U-Net [58]	MobileNetv3-Small 0.75	2.88	11.04	9.82	87.31	93.51	92.94	93.22
ERFNet [64]	-	2.06	7.9	14.73	87.58	93.18	93.58	93.38
MFNet [66]	-	1.34	5.12	4.60	89.04	94.44	93.97	94.2
FastSCNN [67]	-	**1.14**	**4.42**	**0.88**	85.34	92.61	91.57	92.09
small								
LETNet [80]	-	0.95	3.71	7.3	88.16	94	93.41	93.71
ContextNet [68]	-	0.87	3.42	0.89	83.84	91.48	90.94	91.22
DABNet [69]	-	0.75	2.9	5.28	87.64	93.18	93.65	93.41
EDANet [70]	-	0.68	2.61	4.45	87.5	93.09	93.57	93.33
SGCPNet [78]	MobileNetv3-Small 1.0	0.61	2.37	0.44	84.69	92.4	91.03	91.71
CFPNet [71]	-	0.55	2.12	4.00	87.01	93.37	92.74	93.05
CGNet [72]	-	0.49	1.92	3.55	88.22	94.16	93.32	93.74
ENet [73]	-	0.36	1.39	2.84	87.09	93.04	93.16	93.1
FSSNet [74]	-	0.17	0.68	1.37	84.64	91.81	91.56	91.68
FPENet [75]	-	**0.11**	**0.46**	**0.78**	84.45	91.81	91.34	91.57
DASK-Net(ours)	-	0.48	1.91	4.85	**89.54**	**94.76**	**94.21**	**94.48**

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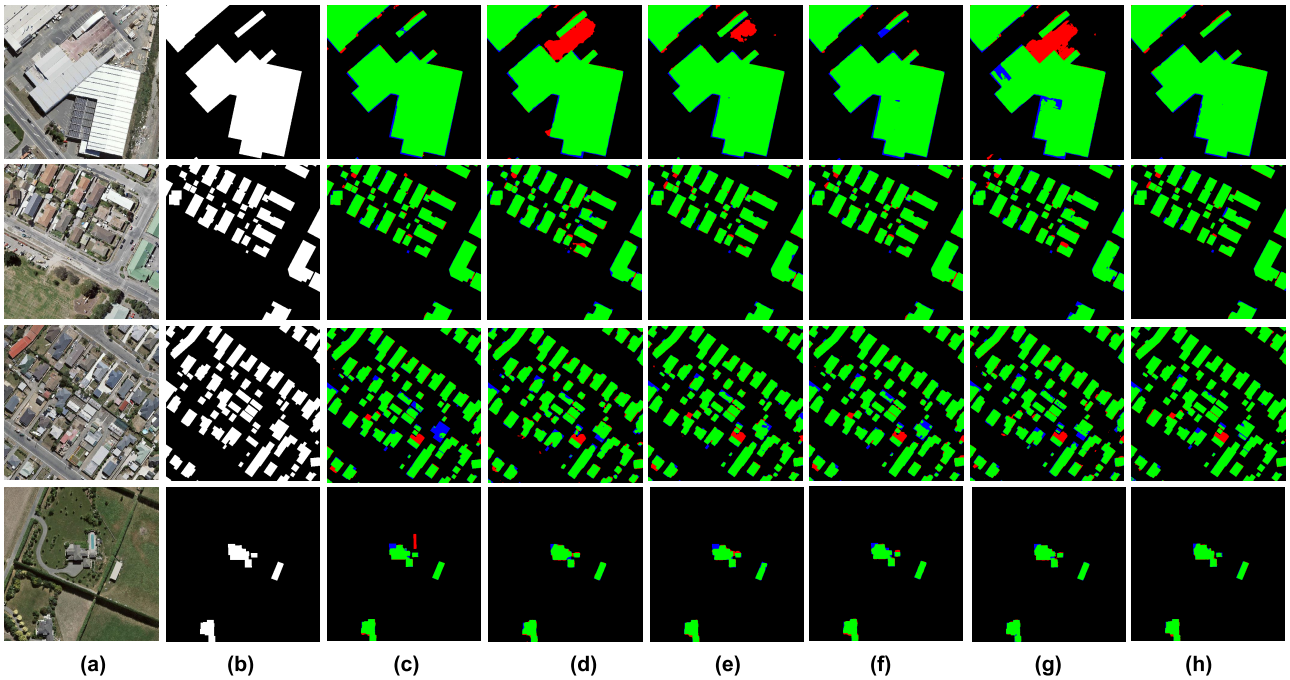


Fig. 8. Visual comparison on the WHU building dataset. (a) Original image. (b) Original segmentation mask. (c) U-Net. (d) FastSCNN. (e) DABNet. (f) CGNet. (g) FPENet. (h) DASK-Net(ours). False positive pixels are marked in red, false negative pixels are marked in blue, and true positive pixels are marked in green.

D. Results on WHU Building Dataset

The quantitative results in Table IV demonstrate that our proposed DASK-Net attains an $F1$ score of 94.48%, a precision of 94.76%, and an IoU of 89.54% while maintaining minimal computational requirements. The results of models with * are reported from the original paper.

In the large model category, low-level feature enhancement and multiscale attention pyramid aggregation network (LFEMAP-Net) [81] achieves the highest $F1$ score (95.34%), but it requires 127.37M parameters—over 265 times more than DASK-Net. Similarly, BuildFormer [15] delivers strong performance ($F1$: 95.06%) but at the cost of 40.52M parameters and 116.58 GMACs.

Medium models like DeepLabv3+ [64] ($F1$: 94.29%) and EasyNet [42] ($F1$: 94.2%) provide a better efficiency-accuracy trade-off with 5.22M and 3.9M parameters, respectively. However, DASK-Net matches or exceeds their performance despite using significantly fewer parameters.

Among small models, DASK-Net substantially outperforms all competitors. The next best small model, LETNet [78], achieves an $F1$ score of 93.71% with nearly twice the parameter count (0.95M). Notably, DASK-Net's performance rivals several medium and large architectures while requiring orders of magnitude fewer resources. These experimental results demonstrate that the proposed method can achieve state-of-the-art performance in the building footprint extraction task while maintaining a lightweight architecture.

Fig. 8 visually compares several benchmark methods and the proposed DASK-Net on the WHU building dataset test set. The images are categorized from top to bottom as large buildings, medium-sized buildings, small buildings, and rural area buildings. Fig. 8 shows that DASK-Net consistently outperforms all benchmark methods across all image types, producing segmentation maps that closely resemble the original masks.

DASK-Net's exceptional performance is primarily due to its utilization of varying kernel sizes and sequential attention mechanisms. These features enable the network to accurately detect even tiny buildings in images, as demonstrated in the second and third images from the top. Unlike benchmark models, DASK-Net maintains the rectangular shape and acute angles of buildings in its segmentation maps. In contrast, benchmark models often fail to accurately segment these features, resulting in oval-like shapes or the omission of acute angles, as seen in the second, third, and fourth images from the top.

Moreover, DASK-Net exhibits superior robustness to occlusions, such as trees, as illustrated in the fourth image. It also demonstrates improved capability in differentiating between building-like objects, such as cars and containers, as shown in the first image. This enhanced capability is particularly valuable in complex urban environments, where multiple object types may share similar visual characteristics.

DASK-Net's visual results highlight its effectiveness in addressing critical challenges in building segmentation. These include handling scale variation, preserving shapes, managing occlusion, and distinguishing between objects. Consequently, DASK-Net substantially advances the state-of-the-art in this domain.

E. Ablation Studies

This section presents a comprehensive ablation study on the WHU building dataset to validate our design choices. We ensure consistent experimental settings with benchmark methods throughout our analysis. Our study is structured to

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TABLE V
INDIVIDUAL CONTRIBUTION OF THE PROPOSED ATTENTION MODULES

Methods	Params(M)	MACs(G)	IoU	F1
EI	**0.44**	**4.77**	86.44	92.73
SAM	0.86	5.55	87.25	93.19
CAM	0.92	5.55	88.41	93.85
EI + SAM	0.46	4.85	87.63	93.41
EI + CAM	0.48	4.85	88.1	93.68
DASK	0.92	5.56	**88.55**	**93.92**

provide insights into the effectiveness of each component and the synergistic benefits of our overall design.

We began by evaluating the individual contribution of each attention module, integrating them separately into the network architecture to assess their impact on performance. Next, we investigated the effects of coupling the attention modules and explored their optimal ordering within the network. Furthermore, we analyzed the computation-performance trade-offs of DASK-Net across different channel configurations. Finally, we incorporated state-of-the-art attention mechanisms into our architecture to compare the performance of our proposed DASK module against them. These experiments are designed to elucidate the specific contributions of each component and demonstrate the rationale behind our architectural decisions. Sections IV-E1–IV-E4 detail our findings from each stage of the ablation study, providing a clear understanding of how our design choices enhance the overall effectiveness of the proposed model.

1) Individual Integration of SAM, CAM, and DASK Into the Encoder Stages: Table V presents the experimental results of incorporating individual attention modules into the architecture. Our baseline, using only EI units throughout the encoder, achieves reasonable performance with minimal parameters. When we replace all encoder layers with individual attention modules, a dimensional mismatch occurs in the first layer of the encoder stages. Since the attention modules (SAM, CAM, DASK) process input of channel size C but output $2C$, an additional 1×1 convolution followed by batch normalization and ReLU layers is applied to the skip connection to match dimensions before summation. Replacing all encoder layers with SAM showed moderate improvements, confirming that spatial context modeling enhances feature representation. The CAM configuration demonstrated more substantial gains than SAM, highlighting the importance of channel-wise feature recalibration for segmentation tasks.

In hybrid approaches, EI + SAM (using EI in the first layer and SAM in the second layer of each encoder stage) achieved better performance than the baseline while using fewer parameters than the implementation of full SAM, demonstrating an efficient way to incorporate spatial attention. Similarly, EI + CAM followed this pattern, but it delivered stronger improvements, nearly matching the full CAM implementation with greater efficiency.

The complete DASK configuration, which integrates both attention mechanisms throughout the encoder, achieved the best overall performance. This validates our dual-attention strategy's effectiveness in capturing complementary feature relationships while maintaining reasonable computational requirements.

TABLE VI
COMBINATIONS OF CAM AND SAM

Methods	Params(M)	MACs(G)	IoU	F1
EI + CAM & SAM	0.48	4.85	88.5	93.9
EI + CAM + SAM	0.48	4.85	88.88	94.11
EI + SAM + CAM	0.48	4.85	**89.54**	**94.48**

2) Coupling SAM and CAM and Their Ordering Structure: Table VI illustrates the results of our experiments with various configurations of spatial and channel attention within the DASK module. Given the distinct impact of each module on the model's overall performance, we investigated how their arrangement affects the outcomes. Initially, we implemented a parallel configuration, applying channel and spatial attention concurrently within the DASK module. In this setup, we fused the final attention maps before the expansion layer, applied the expansion layer to the fused map, and then performed a summation operation to combine the input with the final attention map, generating the final output of the parallelly combined DASK module.

Subsequently, we explored sequential arrangements of these attention mechanisms. Our findings indicate that generating attention maps sequentially yields a more refined attention map than parallel processing. This sequential approach allows one type of attention to guide the other, resulting in better feature refinement. Notably, the spatial-first order demonstrated a slight performance advantage over the channel-first configuration.

The sequential integration of SAM and CAM within our DASK framework enables effective capture of both global contextual relationships and local spatial details. Based on these insights, we adopted the spatial-first sequential order as the optimal design for our DASK module. We incorporated it into the architecture to propose the final DASK-Net model. This configuration effectively leverages the strengths of both attention mechanisms, enhancing the model's overall efficacy in generating accurate and robust segmentation maps from remote sensing imagery.

3) Computation-Performance Tradeoff of DASK-Net: Our extensive evaluation of computation-performance tradeoffs, presented in Table VII, demonstrates the efficiency of DASK-Net across different channel configurations. By systematically increasing channel sizes ($C = 8$, 16, 24, 32, and 48), where C represents the number of feature maps (channels) output by the EI and DASK layers at the first encoder stage and doubles at each subsequent stage, we observed consistent performance improvements up to $C = 32$ across all datasets. Interestingly, further increasing to $C = 48$ yielded dataset-dependent behavior: performance slightly decreased on the Massachusetts Roads dataset while continuing to improve marginally on DeepGlobe and WHU Building datasets. This came at the cost of more than doubling the parameter count (from 0.48M to 1.02M) and substantially increasing MACs (from 4.85G to 9.64G). These results validate our choice of $C = 32$ as the optimal configuration, offering the best balance between computational efficiency and segmentation accuracy for practical remote sensing applications, where resource constraints often play a crucial role in model deployment.

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TABLE VII
COMPUTATION-PERFORMANCE TRADE-OFF OF DASK-NET ACROSS MULTIPLE DATASETS

Channels	Model Complexity		Massachusetts Roads		DeepGlobe		WHU Building	
	Params(M)	MACs(G)	IoU	F1	IoU	F1	IoU	F1
$C = 8$	**0.04**	**0.19**	62.79	77.15	59.48	74.59	84.55	91.63
$C = 16$	0.14	0.56	63.59	77.74	63.37	77.57	85.95	92.45
$C = 24$	0.29	1.13	64.64	78.52	64.95	78.75	87.51	93.34
$C = 32$	0.48	4.85	**65.47**	**79.13**	65.54	79.18	89.57	94.50
$C = 48$	1.02	9.64	65.13	78.82	**65.70**	**79.30**	**90.29**	**94.90**

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART (SOTA) ATTENTION MODULES

Methods	Params(M)	MACs(G)	IoU	F1
EI only	0.44	4.77	86.44	92.73
EI + CBAM	**0.39**	**3.58**	87.86	93.54
EI + SK	0.62	5.78	88.75	94.04
EI + DASK	0.48	4.85	**89.54**	**94.48**

4) Comparison of SK, CBAM, and the Proposed DASK Module: To further validate the effectiveness and superiority of our proposed DASK module, we conducted comparative experiments by integrating state-of-the-art attention mechanisms into our DASK-Net architecture. Specifically, we tested SKs [48] and CBAM [45]. For a fair comparison, we integrated both SK and CBAM into the second part of the encoder stages, mirroring the placement of our DASK module. Furthermore, for SK, we maintained the same number of branches and kernel sizes as in our DASK module.

Table VIII summarizes the results of these experiments. Integrating CBAM [45] into the architecture slightly reduced the complexity of the model compared to our baseline model, which used only EI units, while improving performance through its compact channel and spatial attention mechanisms.

The integration of the SK module [48], substantially improved performance, but at the cost of increasing the complexity of the model. These results demonstrate that the channel-only attention method, enriched with adaptive receptive fields (SK), can yield finer segmentation maps than combined channel-spatial attention with a single receptive field (CBAM), particularly for pixel-wise dense classification tasks in remote sensing imagery. Our DASK module improved these results by combining adaptive multiscale feature extraction with a spatial-channel mechanism. This approach addresses the multiscale limitations of CBAM [45] and the channel-only focus of SK, resulting in enhanced performance with efficient parameter usage.

These experiments highlight the effectiveness and superiority of our proposed DASK module in terms of both performance and model complexity, showcasing its ability to extract richer channel-wise and spatial-wise feature maps. DASK's adaptive receptive field property enhances its capability to handle the diverse scales and complexities present in remote sensing imagery, resulting in more accurate and robust building extraction across various landscapes.

V. CONCLUSION AND FUTURE WORK

This study addresses the challenge of efficient pixel-wise dense prediction for remote sensing imagery, particularly for deployment on resource-constrained devices such as UAVs. To tackle this issue, we first introduced the EI unit, designed for efficient multiscale feature extraction. Next, we developed the DASK module, which integrates the EI unit with a sequential dual-attention mechanism to capture spatial features through adaptive receptive fields while focusing on important features across both channel and spatial locations. Finally, we incorporated these novel components into a U-shaped encoder–decoder framework to propose the lightweight DASK-Net model.

We conducted extensive experiments on Massachusetts roads, DeepGlobe, and WHU building datasets to validate the effectiveness of the proposed method. The experimental results demonstrate that the proposed DASK-Net achieves competitive performance in road network extraction and superior performance in building extraction tasks while substantially reducing computational complexity compared to state-of-the-art models. On the Massachusetts roads and DeepGlobe dataset, DASK-Net achieves accuracy comparable to MSMDFFNet while requiring over 80 times fewer parameters. For the WHU building dataset, DASK-Net demonstrates superior performance compared to most sophisticated models and surpasses all lightweight architectures in terms of IoU and $F1$ scores while maintaining exceptional computational efficiency.

Beyond road and building extraction tasks, DASK-Net has significant potential for broader remote sensing applications. The model's efficiency makes it particularly valuable for disaster response scenarios, where it could enable real-time segmentation of post-disaster imagery to guide emergency operations. In our future work, we will evaluate DASK-Net on disaster-specific datasets and investigate its robustness to challenging conditions typical in emergency contexts. We also plan to explore integration with multimodal data streams to enhance damage detection capabilities. Additionally, DASK-Net shows promise for environmental monitoring applications such as deforestation tracking and urban development analysis, where its lightweight nature facilitates continuous monitoring via UAV deployments. These expanded applications represent promising directions for extending the impact of our lightweight segmentation approach beyond its current implementation.

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